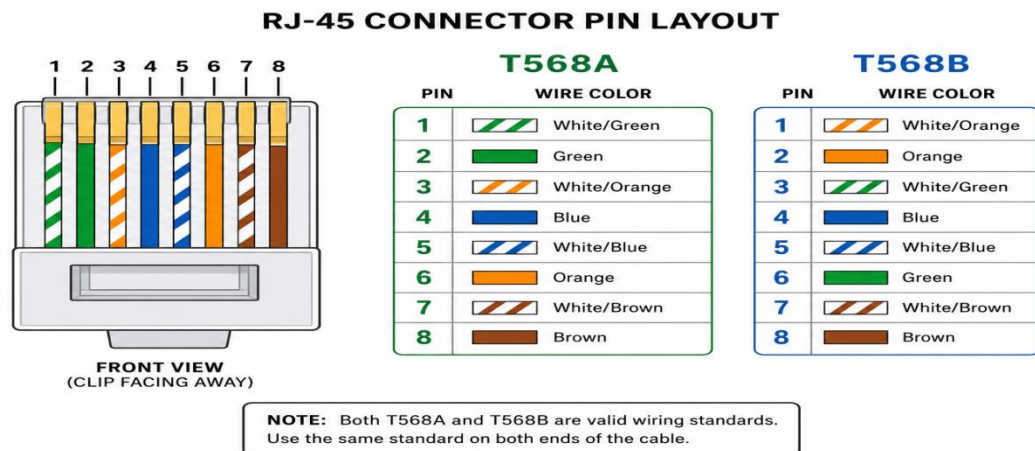


1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100Mbps	1000Mbps
Cable	Cat3 or higher UTP	Cat5 or higher UTP	Cat5e or Cat6 UTP
Maximum cable length	100 meters	100 meters	100 meters
Connector type	RJ-45	RJ-45	RJ-45

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

- To connect two laptops directly using Ethernet. It requires an Ethernet crossover cable.
- In Ethernet, each device transmits (TX) data on one pair of wires and receives (RX) data on another.
- Two laptops are similar devices, so their TX pin connects to TX pins and RX pin connects to RX pins. If you use a straight-through cable. As a result, they cannot communicate properly.

- A crossover cable swaps the transmit and receive wire pairs. This allows the two laptops send and receive data directly.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

a) Desktop PC

- Home or college computer for study, gaming or programming.
- Likely Ethernet standard : 1000BASE-T (Gigabit Ethernet)
- Maximum speed : 1 Gbps

b) Wi-Fi Router

- Connects the home or college network to the internet.
- Likely Ethernet standard : 1000BASE-T (LAN ports on modern routers)
- Maximum speed : 1 Gbps

c) Network Printer

- Shared printer in a home office or college lab.
- Likely Ethernet standard : 100BASE-TX (Fast Ethernet)
- Maximum speed : 100 Mbps